Intro to Machine Learning
Practice Problems 1: Bias, Variance and the Curse of Dimensionality

Practice problems, not turned in

Directions: These practice problems go with Week 1 (lecture: *What Is Machine Learning? Bias, Variance, and the Curse of Dimensionality*). They are not turned in, but the quizzes draw on them, so work through each one before looking at the solutions. Each problem is adapted from the exercise cited next to it; see the book for the original wording.

1. **Flexible or inflexible?** (ISL 2.4, Exercise 1; ISL 2.1.3, 2.2.2.) For each situation, say whether you would generally expect a *flexible* learning method to perform better or worse than an *inflexible* one, and justify your answer using bias and variance.

   (a) The sample size n is very large and the number of predictors p is small.

   (b) The number of predictors p is very large and n is small.

   (c) The relationship between the predictors and the response is highly non-linear.

   (d) The error variance $\sigma^2 = \mathrm{Var}(\varepsilon)$ is very large.

2. **Regression or classification? Inference or prediction?** (ISL 2.4, Exercise 2; ISL 2.1.1, 2.1.5.) For each scenario, say whether it is a regression or a classification problem, whether the main interest is inference or prediction, and give n and p.

   (a) For each of the 500 largest US firms we record profit, number of employees, industry and CEO salary, and we want to understand which factors affect CEO salary.

   (b) Before launching a new product, we want to know whether it will succeed or fail. We have 20 similar past products, and for each we know whether it succeeded, its price, its marketing budget, the competitor's price and ten other variables.

   (c) We want to predict the weekly % change in the USD/Euro exchange rate from the weekly % changes in the US, British and German stock markets, using weekly data for all of 2012.

3. **The bias–variance picture.** (ISL 2.4, Exercise 3; ISL 2.2.2; lecture §4.)

   (a) On a single plot, sketch typical curves for squared bias, variance, training error, test error and irreducible (Bayes) error as a method goes from less flexible to more flexible. Put flexibility on the x-axis and label all five curves.

   (b) Explain why each curve has the shape you drew.

4. **Flexibility: pros and cons.** (ISL 2.4, Exercise 5; ISL 2.1.3.) What are the advantages and disadvantages of a very flexible approach to regression or classification, compared with a less flexible one? When would you prefer each?

5. **KNN by hand.** (ISL 2.4, Exercise 7; ISL 2.2.3.) The training set below has six observations, three predictors and a qualitative response. We want to predict Y at the test point $X_1 = X_2 = X_3 = 0$ using K-nearest neighbours with Euclidean distance.

Obs.	X_1	X_2	X_3	Y
1	0	3	0	Red
2	2	0	0	Red
3	0	1	3	Red
4	0	1	2	Green
5	-1	0	1	Green
6	1	1	1	Red

   (a) Compute the Euclidean distance from each observation to the test point.

   (b) What is the prediction with $K = 1$? Why?

   (c) What is the prediction with $K = 3$? Why?

   (d) If the Bayes decision boundary is highly non-linear, would you expect the best K to be large or small? Why?

6. **The curse of dimensionality for KNN.** (ISL 4.8, Exercise 4; lecture §5.1.) Observations are uniformly distributed on $[0, 1]$ in each feature. To predict at a test point, we use only training observations within the 10% of each feature's range closest to that test point. For example, with $p = 1$ and test point $X = 0.6$, we use observations with $X \in [0.55, 0.65]$.

   (a) With $p = 1$, on average what fraction of the observations do we use?

   (b) With $p = 2$, and (X_1, X_2) uniform on $[0, 1]^2$, we use observations within 10% of the range of X_1 *and* within 10% of the range of X_2. What fraction do we use?

   (c) What fraction do we use with $p = 100$?

   (d) Use (a)–(c) to argue that when p is large, very few training observations are "near" any test point.

   (e) Now build a p-dimensional hypercube centred on the test point that contains, on average, 10% of the training observations. For $p = 1$, 2 and 100, how long is each side? Comment.